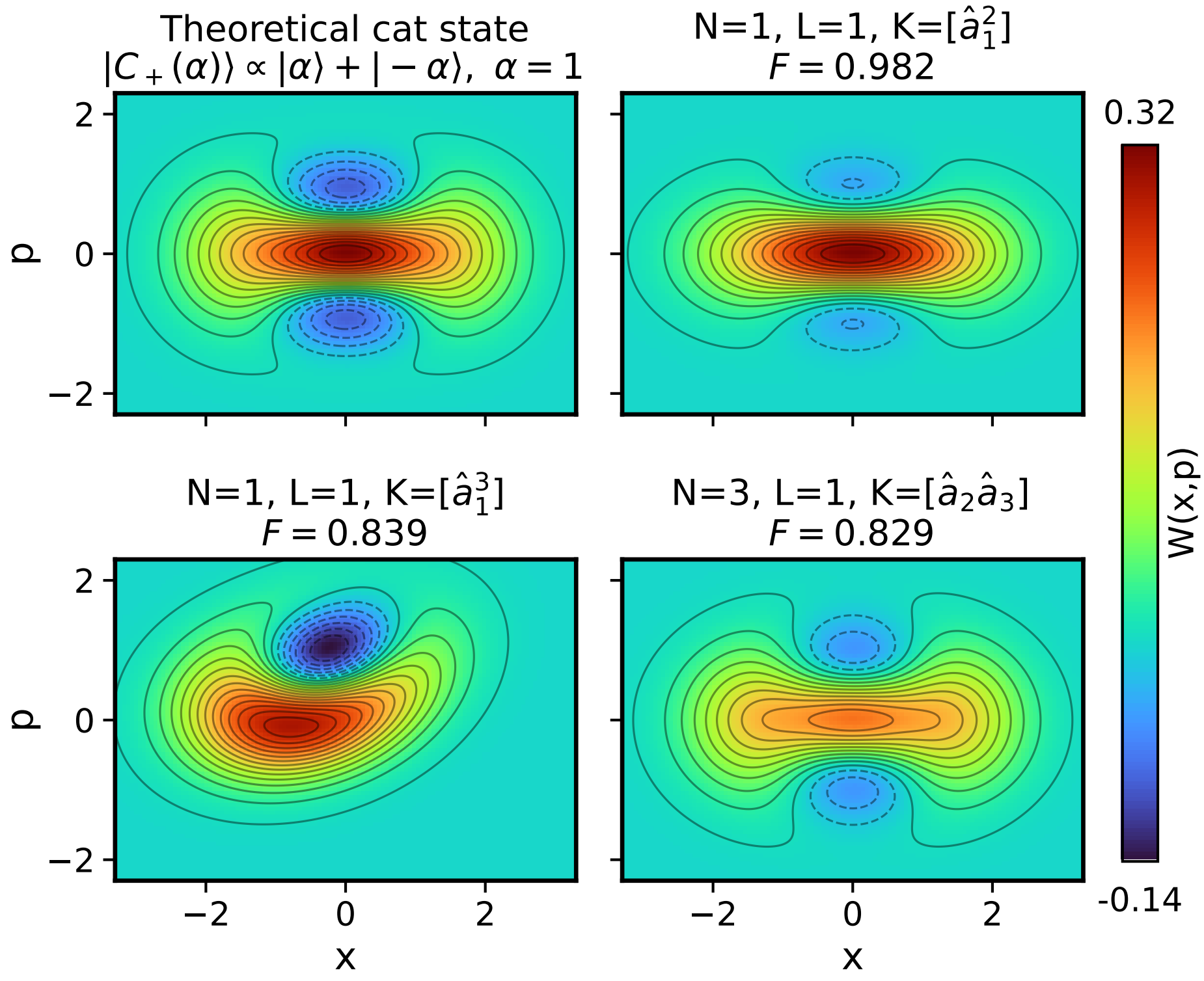

Theoretical cat state
$|C_+(\alpha)\rangle \propto |\alpha\rangle + |-\alpha\rangle,\ \alpha = 1$
N=1, L=1, K=$[\hat{a}_1^2]$
$F = 0.982$
N=1, L=1, K=$[\hat{a}_1^3]$
$F = 0.839$
N=3, L=1, K=$[\hat{a}_2\hat{a}_3]$
$F = 0.829$
p
x
W(x,p)
0.32
-0.14
2
0
−2